

CarthaPOS

Intelligent SaaS platform for automated POS generation
and Business Intelligence

Software Engineering · Business Intelligence

STUDENT

Oussama Ferjani

ACADEMIC YEAR

2025 – 2026

COMPANY

MAKTABI SARL

ACADEMIC SUPERVISOR

Mme Safa Fennia



The Problem

Traditional POS software is built once and rarely bends

TRADITIONAL POS



- ✓ Fixed products & features
- ✓ Hard to customize
- ✓ Separate, manual deployments
- ✓ No centralized control
- ✓ Weak Business Intelligence link

THE BUSINESS NEED

How can we generate customized POS systems for different business sectors — while managing clients, licences, modules and analytics from a single platform?

SIX BUSINESS SECTORS, ONE GENERATOR

Restaurants

Cafés

Bakeries

Retail

Pharmacies

Salons

Existing Solutions

Three mature categories — none covers the full need

CATEGORY 1

Traditional POS

- Sector-specific, but fixed
- No central client management
- No analytics layer

CATEGORY 2

SaaS admin platforms

- Manage accounts & billing
- Do not build the POS software
- Generic, not sector-aware

CATEGORY 3

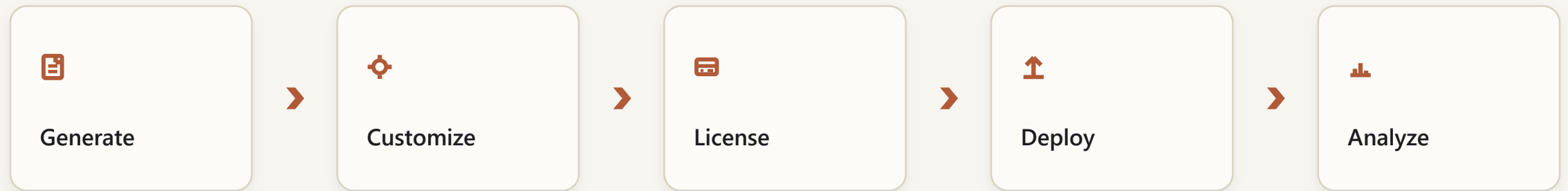
Standalone BI tools

- Powerful dashboards
- No POS data model
- No per-client isolation

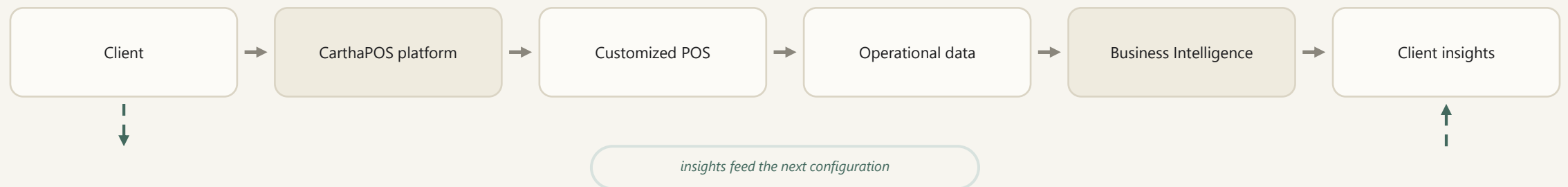
CarthaPOS = the intersection of all three

CarthaPOS

One continuous lifecycle — from a client to actionable insight



THE ECOSYSTEM



System Context

Global use-case view — actors, the platform boundary and external services

UML diagram area

◇ Insert your existing UML diagram here, then delete this note

Global Class Architecture

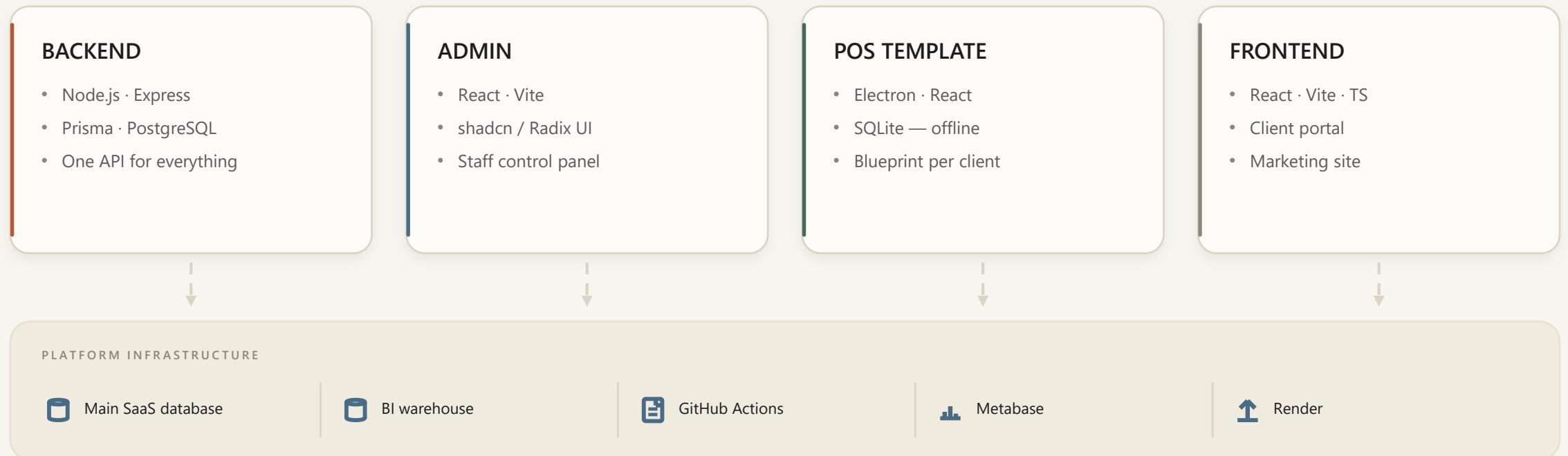
Core domain model — clients, licences, modules, POS projects and BI entities

UML diagram area

◇ Insert your existing UML diagram here, then delete this note

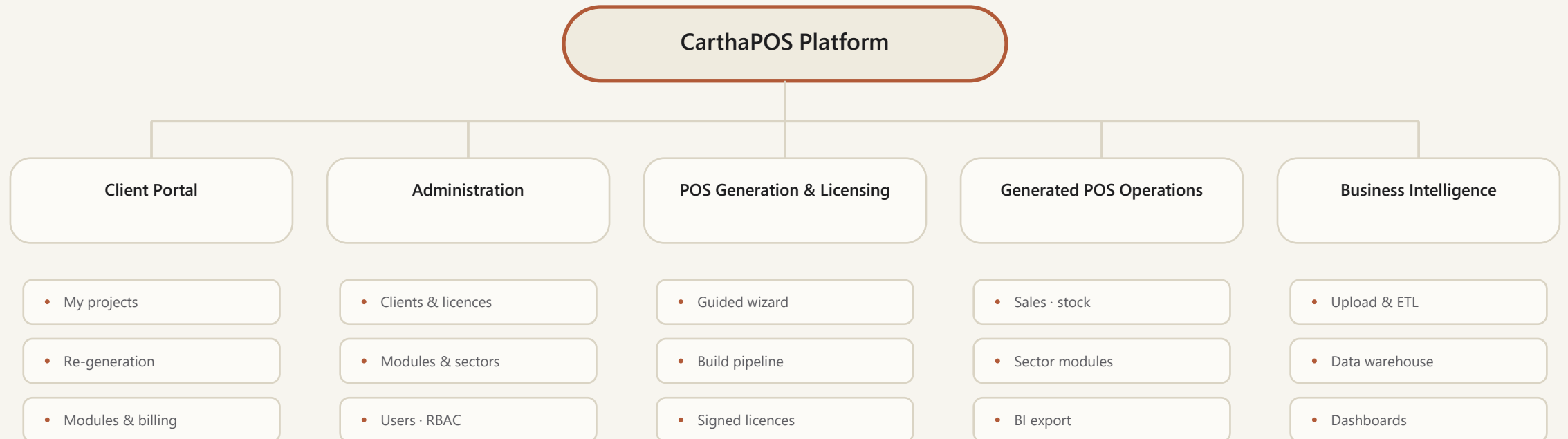
Technical Architecture

Four independently deployed projects around one API



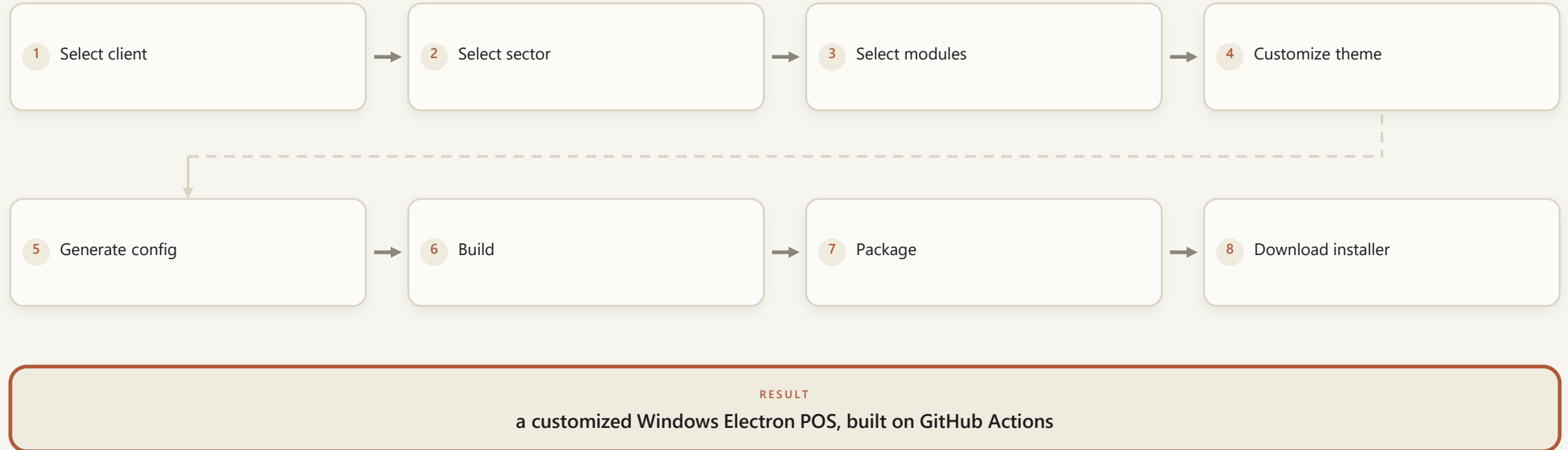
Platform Functional Architecture

Five functional domains served by one shared backend



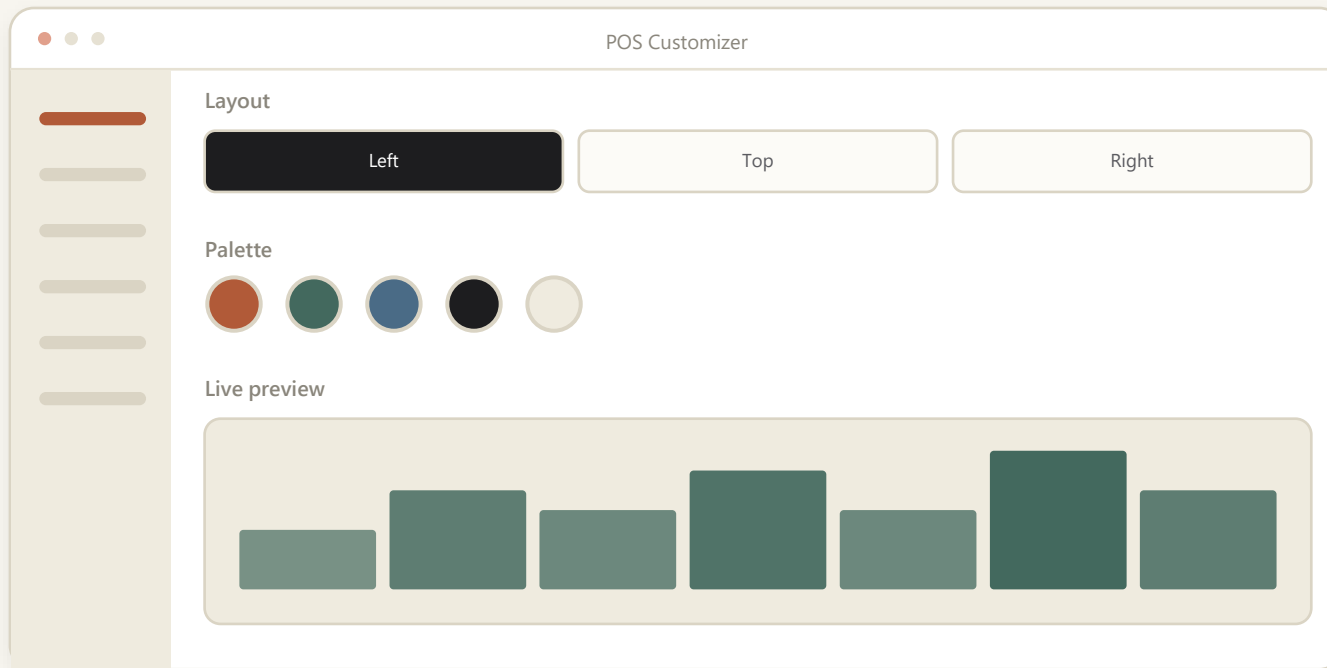
From Configuration to POS

A guided pipeline turns a few choices into a Windows installer



Customize the POS

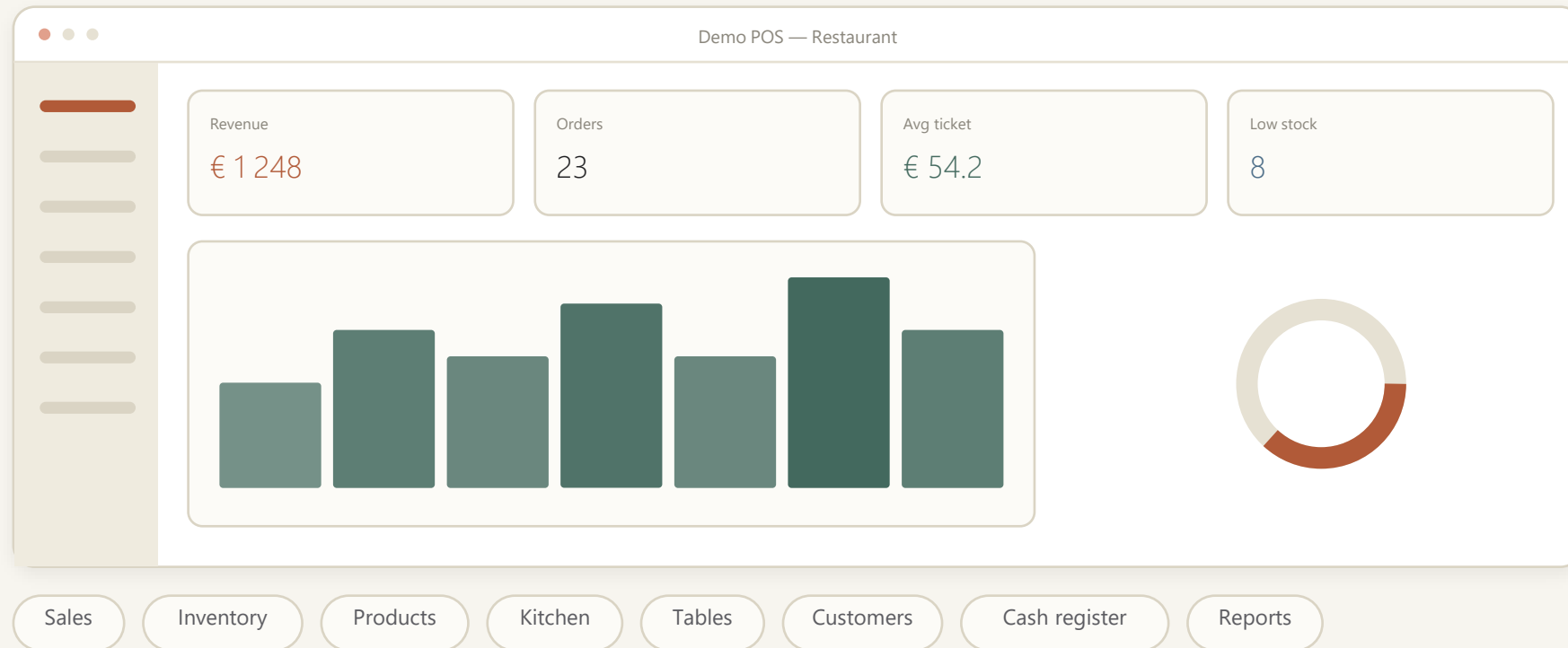
A live customizer — every choice is previewed before generation



- Business sector
- Modules
- Colours & logo
- Layout & navigation
- POS configuration

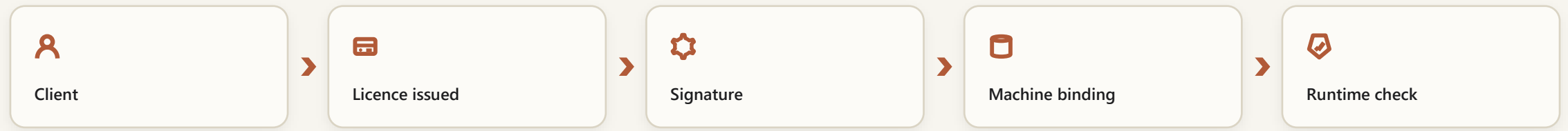
Generated POS Application

An offline-capable Electron desktop app, tailored to the sector

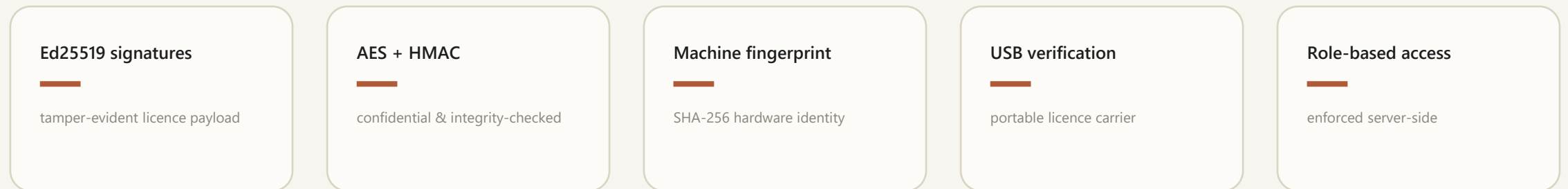


Security & Licensing

Every generated POS is cryptographically bound and verified at runtime



HOW IT IS PROTECTED



Role-Based Access Control

A single permission model gates both the API and the interface



ROLES

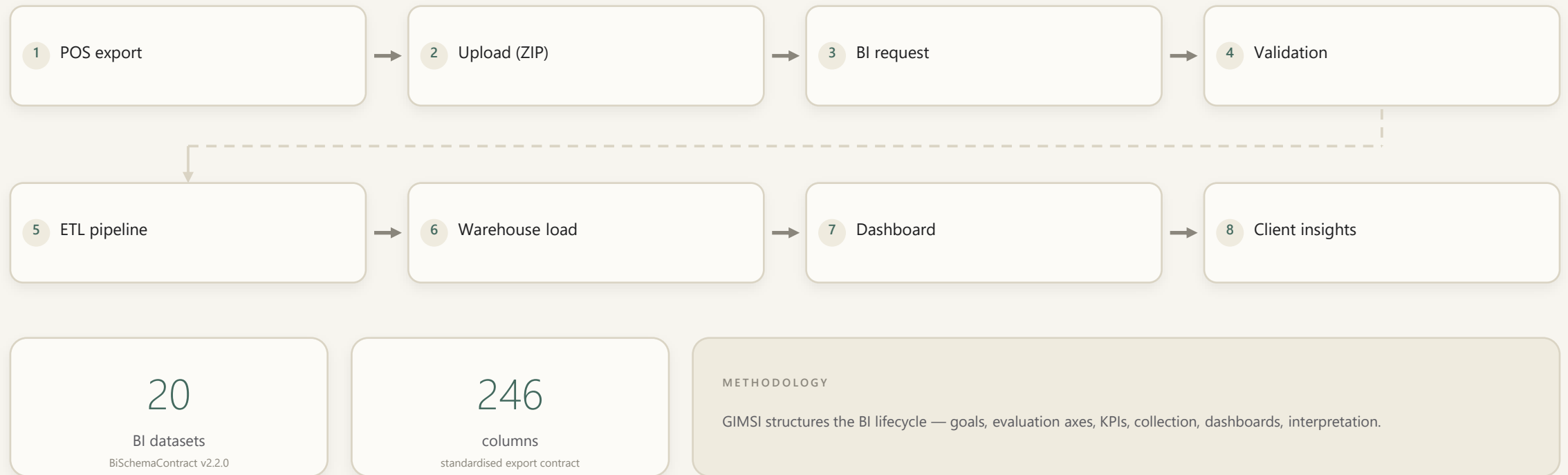
- SUPER_ADMIN
- ADMIN
- MANAGER
- BI_SPECIALIST

35
permission keys

10
permission groups

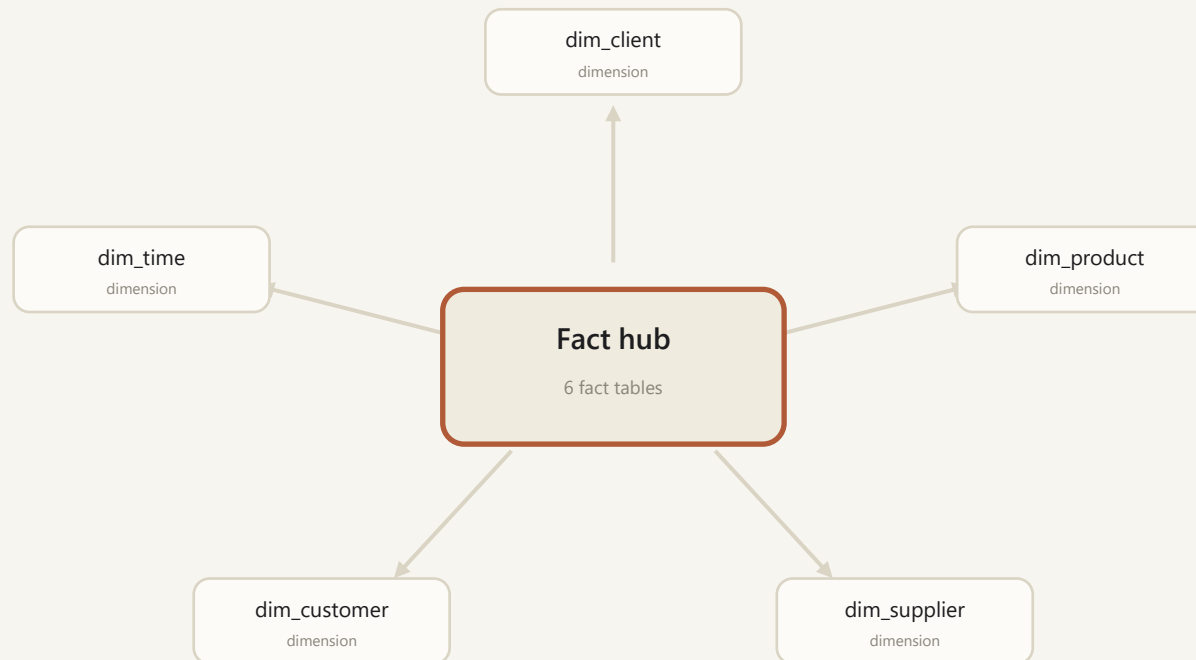
From POS Data to Business Intelligence

A governed pipeline turns raw exports into per-client dashboards



Analytical Data Warehouse

A tenant-isolated star schema in PostgreSQL



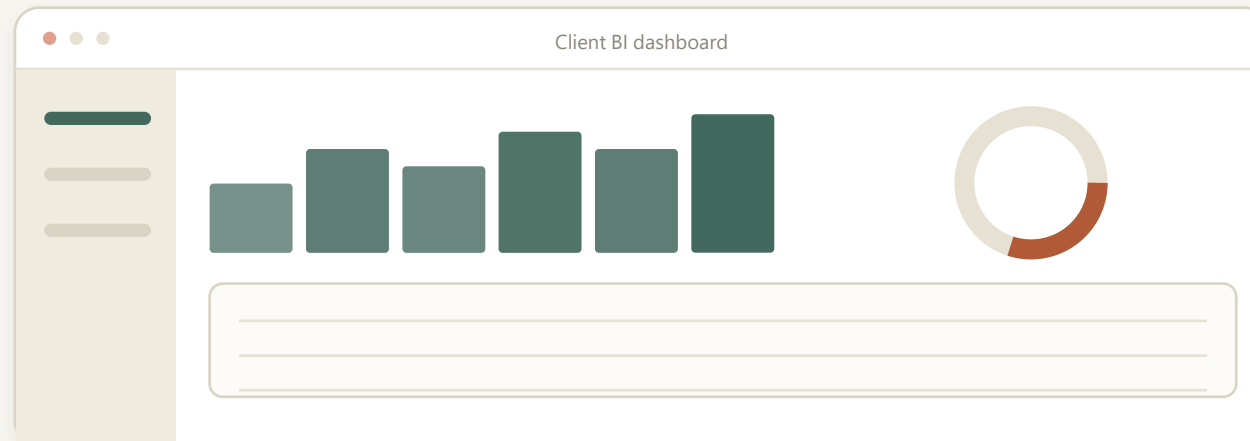
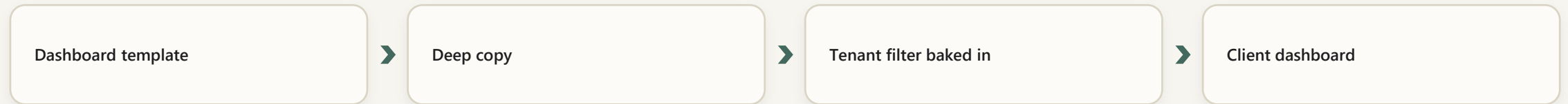
FACT TABLES

- fact_sales
- fact_sale_items
- fact_inventory
- fact_appointments
- fact_kitchen_orders
- fact_kitchen_order_items

- Multi-tenant — every row carries tenantId
- Conformed dim_time across all facts
- Idempotent ETL — safe to re-run

Personalized BI Dashboards

Dashboards are provisioned per client, never shared



Each client sees only their own analytical data.

Metabase CE has no row-level security — isolation is baked into each copied dashboard at provisioning time.

Build & Deployment Pipeline

From source control to production, with automated Windows builds



KEY PRACTICES

Automated POS installer builds

Docker containerization

PostgreSQL (managed)

Static admin & frontend

CI/CD via render.yaml

DEPLOYMENT TOPOLOGY

- carthapos-backend
- carthapos-db
- Docker web service
- managed PostgreSQL
- admin + frontend
- Windows installer
- static sites
- GitHub Actions runner

Results

What the platform delivers today

6

business sectors

20

BI datasets

246

BI columns

35

RBAC permissions

6

warehouse facts

5

warehouse dimensions

Fully automated Windows POS generation, end to end

Limitations & Future Perspectives

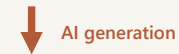
LIMITATIONS

- Local Metabase & single-region deployment
- ETL runs synchronously on upload — no job queue
- Per-tenant Metabase sandboxing is still manual
- Some advanced modules exist only as previews

FUTURE VISION

PROMPT

"Create a restaurant POS with table management, kitchen orders, loyalty and dark-blue branding."



Customized POS, generated from language

ALSO ON THE ROADMAP

Full ERP

AI-assisted config

Dedicated POS hardware

Smarter analytics

Automation

CONCLUSION

CarthaPOS



One platform connecting POS generation, centralized management and Business Intelligence.

Thank you